Figure 3: The LVM configuration tool

Figure 4: Initialising a disk/entity

Setting up LVM during Install

Ubuntu's default installation image does not allow LVM, so we will use the Ubuntu Alternate Install disc. Grab the alternate ISO from the Ubuntu Download page (it's under '*Additional options*') and boot from it. I would like to go on record to state that I was able to do all the steps below using the standard ISO of Ubuntu 12.10 as well—which means the alternate ISO is not required, even though Internet sources claim one must use the alternate ISO for LVM support.

On the disk partitioning screen (Figure 2), select '*Erase disk and install Ubuntu*'; check the '*Use LVM with the new Ubuntu installation*' and click '*Continue*'. After selecting the LVM partitioning option on the previous screen, the installer will automatically select the main hard drive to use for initialising the LVM set-up. The installer will then write a basic layout to disk. That's about it. The installer now installs GRUB—which happens in the MBR, since the BIOS can't read LVMs. Any extra */boot* partition required will automatically be created, and the installer will then proceed with the rest of the installation.

It's super easy, isn't it? Now, using LVM tools, you can configure your LVM to add or remove more disks, extend or shrink volumes, etc.

Setting up or configuring LVMs in an existing system

I will show you how to configure or modify the previously created volumes. The same steps may be used to set up a fresh LVM configuration if you have Ubuntu already installed.

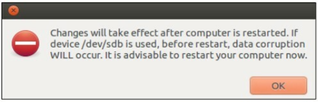

Figure 5: Initialising a disk/entity

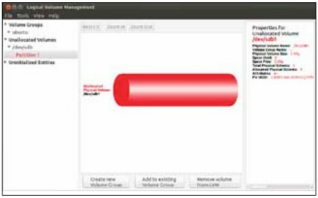

Figure 6: Initialising a disk/entity

Figure 7: Unused space now available in the existing VG

Let's install the GUI tool *'system-config-lvm'*. Initially available only from the Fedora RPM, this tool is now supported from the Ubuntu repos, thus saving you a lot of work including 'transcoding' the RPM to DEB using *alien*. To install it is now as simple as issuing the following command at a terminal:

```
sudo apt-get update
sudo apt-get install system-config-lvm
```

Once installation is complete, fire it up from the main menu (search for 'LVM'), or you can launch it from a terminal with *sudo system-config-lvm*. The application's main screen is shown in Figure 3.

Next, I will show you how to add a new disk and partition (physical volume) to the volume group, and extend the existing (root) logical volume. So, let's touch upon adding and editing logical volumes.

Adding a new disk/partition to an existing volume group: The uninitialised disk is shown in the '*Uninitialised Entities*' branch; so initialise it by hitting the '*Initialise*